Two-Component Continuation Control in Navier–Stokes:

The Matrix-of-Error Framework and Smoothly Forced Breakdown

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Abstract

The nonlinear-calculus matrix-of-error framework leads, in its three-dimensional Navier–Stokes application, to a continuation coefficient proportional to $\nu^{-3}\|P_\chi\omega\|_{L^2}^4$, where P_χ retains two fixed vorticity components. This note recalls the identification, made in [1], of the condition with the spatial-exponent-two case of the Chae–Choe criterion, explains its origin in the stretching estimate, and compares its unforced use with the smoothly forced breakdown reported by OpenAI.

Three statements organize the comparison. First, interpolation and Young's inequality produce a scale-critical continuation integral that is not controlled by the usual energy bound alone. Second, smooth forcing adds a finite inhomogeneous contribution to the enstrophy envelope without changing the functional form of the blockwise stretching coefficient; it nevertheless changes the evolving field and can change the coefficient's accumulated value. Third, accepting the reported periodic breakdown theorem, the selected two-component integral must diverge at its finite singular horizon. The continuation criterion and that construction are therefore compatible: the construction lies outside the unforced input class and fails the sufficient condition in the enlarged forced class. Exact finite examples and a qualified core-scaling comparison make this relationship explicit. The note does not settle class-wide finiteness for unforced data.

1 The Comparison Question

Nonlinear Calculus and Solving Three-Dimensional Navier–Stokes [1] introduced a cross-system matrix-of-error framework and developed its three-dimensional fluid representation through derivation, numerical validation, and a smooth-continuation interpretation. The present note concentrates on the specific condition at the end of that analysis. What produces the condition, and how can the same condition coexist with a smoothly forced finite-time breakdown?

The comparison is between a representation with a conditional regularity consequence and a constructed family with a reported singular endpoint. The underlying regularity criterion is established mathematics [2]; its selection and connection to the matrix-of-error framework belong to [1]. OpenAI's construction [3] supplies an input family against which the scope of that condition can be made precise.

The earlier framing of this investigation anticipated a landscape of solution families rather than a single construction classifying every admissible history. Recent reported constructions give

concrete examples of that landscape: OpenAI's smoothly forced Navier–Stokes family, Alpöge and Buckmaster's smoothly forced Boussinesq construction [9], and Alpöge, Buckmaster, and Coiculescu's incompressible porous media (IPM) construction [10]. Alpöge and Buckmaster's corresponding forced Euler result is presented in a standalone preprint [11, Theorem 1.1]. These are families for different equations, not multiple Navier–Stokes constructions within one input class. Their comparison must retain the equations, forcing, domains, and regularity assumptions of each; the viscous criterion examined here applies specifically to Navier–Stokes.

The argument below derives the forcing extension and its consequence for the selected integral. It uses the external breakdown theorem as an attributed premise, not as a theorem independently verified by this note. Its original physical motivation is retained in the preparation note; no additional physical-admissibility rule or complexity classification is needed for the mathematical comparison.

2 Why the Two-Component Condition Appears

Let $\Omega_L = (\mathbb{R}/L\mathbb{Z})^3$, $L > 0$, and $\nu > 0$. Consider smooth periodic incompressible velocity u, pressure p, and prescribed force f:

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u + f, \qquad \nabla \cdot u = 0. \tag{2.1}$$

The unforced setting is $f = 0$. Define vorticity $\omega = \nabla \times u$, strain $S = (\nabla u + \nabla u^{\mathsf{T}})/2$, and

$$Y = \|\omega\|_2^2, \qquad \mathcal{B} = \int_{\Omega_L} \omega^{\mathsf{T}} S \omega \, \mathrm{d}x, \qquad \mathcal{C} = \nu \, \|\nabla \omega\|_2^2. \tag{2.2}$$

All spatial norms here are on Ω_L, unless stated otherwise. Y is twice the enstrophy. In the unforced case, the matrix interpretation and its scalar balance are

$$\frac{D\omega}{Dt} = S\omega + \nu \Delta \omega, \qquad \mathfrak{E}_{\mathrm{NS}} = [S\omega \mid \nu \Delta \omega], \qquad \tfrac{1}{2} Y' = \mathcal{B} - \mathcal{C}, \tag{2.3}$$

where $D/Dt = \partial_t + u \cdot \nabla$. The columns retain physical contributions; closure of their governing relation does not require either column to vanish [1, Secs. 2–5].

2.1 The blockwise estimate and its attribution

Fix a unit vector χ, independent of space and time, and set $\eta = P_\chi \omega$, where $P_\chi = I - \chi \otimes \chi$. The integrated stretching estimate is

$$|\mathcal{B}| \le C \, \|\eta\|_2 \, \|\omega\|_4^2. \tag{2.4}$$

Its two-component structure is that of Chae–Choe [2, Theorem 1 and Eqs. (7)–(9)]. A direct periodic explanation also clarifies which contribution is controlled, rather than discarded. Choose an orthonormal component basis with $\chi = e_3$, and put $\zeta = \omega_3$. Then

$$\mathcal{B} = \int_{\Omega_L} \left(\eta^{\mathsf{T}} S \eta + 2\zeta e_3^{\mathsf{T}} S \eta + \zeta^2 \partial_3 u_3 \right) \mathrm{d}x. \tag{2.5}$$

The periodic Biot–Savart identity gives

$$\partial_3 u_3 = -\partial_3 \Delta^{-1} (\partial_1 \eta_2 - \partial_2 \eta_1), \qquad \|\partial_3 u_3\|_2 \le C \, \|\eta\|_2, \qquad \|S\|_4 \le C \, \|\omega\|_4.$$

Here Δ^{-1} acts on mean-zero fields. Hölder's inequality bounds the first two terms of (2.5) by $C\,\|\eta\|_2\,\|S\|_4\,\|\omega\|_4$, and the last by $\|\omega\|_4^2\,\|\partial_3 u_3\|_2$, proving (2.4). The last term is not assumed to vanish. For another fixed χ, the same component argument uses directional derivatives; the lattice geometry is retained in the periodic operator bounds.

Because a periodic curl has zero mean, interpolation and Young's inequality, with conjugate exponents $4/3$ and 4, give, for $0 < \delta < 1$,

$$\begin{aligned} |\mathcal{B}| &\le C\,\|\eta\|_2\,\|\omega\|_2^{1/2}\,\|\nabla\omega\|_2^{3/2} \\ &\le (1-\delta)\nu\,\|\nabla\omega\|_2^2 + K_\delta \nu^{-3}\,\|\eta\|_2^4\,Y. \end{aligned} \tag{2.6}$$

Thus, with $K_\delta > 0$ an analytic estimate constant,

$$g_\chi = K_\delta \nu^{-3}\,\|P_\chi\omega\|_2^4, \qquad Y' + 2\delta\mathcal{C} \le 2g_\chi Y \quad (f=0). \tag{2.7}$$

The constants C (which may change between estimates) and K_δ can be chosen independently of L: the periodic singular-integral bounds and mean-zero interpolation inequality used here are invariant under rescaling the cubic torus.

Chae–Choe's theorem is stated on $\mathbb{R}^3$; the periodic estimate above supplies the corresponding bound used here. Their mixed-norm condition includes $\widetilde{\omega} \in L_t^a L_x^b$ with $2/a + 3/b = 2$, $b > 3/2$. The choice $(a,b) = (4,2)$ is exactly the selected norm. The matrix framing retains this established criterion rather than creating a different regularity condition. Miller [8] extends the fixed-plane criterion on $\mathbb{R}^3$ to planes varying in space and time, with bounded spatial gradient of the unit normal, in the same $L_t^4 L_x^2$ norm; the present comparison uses only fixed planes.

The role of [1] is to connect the $(a,b) = (4,2)$ member of this established criterion to the matrix-of-error representation. Its selected blockwise majorant expresses the retained stretching and diffusion contributions in an enstrophy envelope, with the factor ν^{-3} and fourth power explicit. This organization makes the comparison with forcing direct: Section 3 adds a vorticity-source column and an inhomogeneous envelope term while retaining the functional form of the stretching coefficient. Section 4 then applies this continuation implication to OpenAI's reported periodic family: accepting that breakdown result, the same accumulated two-component control must diverge at its singular horizon.

The condition also has a physical and geometric interpretation [8, Introduction and Theorem 1.2]. In an unforced planar incompressible flow on the periodic domain, vorticity points perpendicular to the plane, vortex stretching vanishes, and nonlinear transport preserves enstrophy while viscosity dissipates it. In three dimensions, incompressibility allows the two projected vorticity components to control the integrated stretching contribution through (2.4). Their critical space-time norm quantifies a departure from planar vorticity structure relevant to regularity. Finite-time breakdown requires the accumulated critical control to diverge. This explains the significance of two components rather than three: the remaining component's contribution is controlled through the projected pair, as the decomposition above shows.

2.2 Critical control versus the available energy bound

Let $[0, T_*)$ be the maximal smooth interval. For finite $T > 0$, write

$$J_\chi(u;T) = \lim_{t \uparrow \min\{T,T_*\}} \int_0^t \|P_\chi\omega(s)\|_2^4\,\mathrm{d}s. \tag{2.8}$$

This nonnegative integral can be finite or infinite. The continuation requirement is finiteness on every finite horizon, not finiteness of the total integral over $[0,\infty)$. It is not a duration cutoff: accumulation may remain finite on every finite interval while growing without bound as the horizon tends to infinity. The unforced energy balance supplies

$$\frac{1}{2}\|u(t)\|_2^2 + \nu\int_0^t Y(s)\,\mathrm{d}s = \frac{1}{2}\|u_0\|_2^2, \qquad \|P_\chi\omega\|_2^4 \le Y^2. \tag{2.9}$$

Control of $\int Y$ alone does not control $\int Y^2$, and consequently does not provide the required bound on the selected block by this estimate. This is the unresolved step in the energy argument, not a defect in the integrating-factor calculation.

The scaling explains the distinction [4]. On $\mathbb{R}^3$, the Navier–Stokes rescaling is $u_\lambda(x,t) = \lambda u(\lambda x, \lambda^2 t)$, with unchanged viscosity. It gives

$$\|u_\lambda(t)\|_2^2 = \lambda^{-1}\left\|u(\lambda^2 t)\right\|_2^2, \qquad \int_0^{T/\lambda^2}\|P_\chi\omega_\lambda(t)\|_2^4\,\mathrm{d}t = \int_0^T \|P_\chi\omega(s)\|_2^4\,\mathrm{d}s. \tag{2.10}$$

The target integral is scale invariant, or critical, while kinetic energy can decrease under concentration. These are whole-space scaling identities, not an assertion that the period L is unchanged by dilation. They describe why the energy-level control is called supercritical. The factor ν^{-3} and fourth power in (2.7) follow specifically from absorbing the gradient factor in (2.6). This explains the condition without a separate hypothesis about the physical meaning of time, and without claiming that every possible analytical method must stop at the same estimate.

3 What Smooth Forcing Changes

Taking the curl of (2.1) adds a third physical contribution:

$$\mathfrak{E}_{\mathrm{NS},f} = [S\omega \mid \nu\Delta\omega \mid \nabla\times f], \qquad \tfrac{1}{2}Y' = \mathcal{B} - \mathcal{C} + \mathcal{I}_f, \qquad \mathcal{I}_f = \int_{\Omega_L}\omega\cdot(\nabla\times f)\,\mathrm{d}x. \tag{3.1}$$

The stretching estimate depends on the instantaneous divergence-free field, so (2.6) keeps the same form. For any fixed $\alpha > 0$,

$$2\mathcal{I}_f \le \alpha Y + \alpha^{-1}\|\nabla\times f\|_2^2, \qquad Y' + 2\delta\mathcal{C} \le (2g_\chi + \alpha)Y + \alpha^{-1}\|\nabla\times f\|_2^2. \tag{3.2}$$

In dimensional variables, α has units of inverse time. Set

$$A(t) = \int_0^t (2g_\chi(s) + \alpha)\,\mathrm{d}s, \qquad F(t) = \|\nabla\times f(t)\|_2^2. \tag{3.3}$$

Multiplying by $e^{-A(t)}$ and integrating gives the explicit envelope

$$\begin{aligned} e^{-A(t)}Y(t) + 2\delta\int_0^t e^{-A(s)}\mathcal{C}(s)\,\mathrm{d}s &\le Y(0) + \alpha^{-1}\int_0^t e^{-A(s)}F(s)\,\mathrm{d}s, \\ Y(t) &\le e^{A(t)}\left[Y(0) + \alpha^{-1}\int_0^t e^{-A(s)}F(s)\,\mathrm{d}s\right]. \end{aligned} \tag{3.4}$$

The additive source is essential for evolution from rest: an unforced multiplicative envelope with $Y(0) = 0$ would otherwise allow no vorticity production.

Proposition 3.1 (Continuation with smooth forcing)**.** *Let u solve* (2.1) *maximally with smooth periodic divergence-free initial data and $\nu > 0$. Suppose f is smooth on a neighborhood of the finite slab $[0, T]$. If $J_\chi(u; T) < \infty$ for a fixed unit direction χ, then the solution continues smoothly beyond T.*

Proof. The hypothesis bounds A before $\min\{T, T_*\}$. Smooth forcing on the compact periodic slab gives $\int_0^T F(s)\,\mathrm{d}s < \infty$, so (3.4) bounds Y uniformly there. The velocity energy estimate also gives

$$\|u(t)\|_2 \le \|u_0\|_2 + \int_0^t \|f(s)\|_2 \,\mathrm{d}s.$$

It follows by regularizing the norm if it vanishes initially. For periodic divergence-free velocity, $\|\nabla u\|_2^2 = Y$, so u has a uniform H^1 bound. Local existence, uniqueness, and regularity in periodic H^1, with the smooth prescribed force bounded near the slab, permit a restart on a uniformly positive interval from these bounded states [6, Theorem 5.1]. Nonzero spatial means can be removed by the time-dependent translation and velocity shift in [6, Eq. (36)]; smooth forcing bounds those shifts on the slab. Restarting sufficiently close to any putative $T_* \le T$ contradicts maximality, giving smooth continuation. □

This is the precise sense in which the control formula is compatible with both input classes. The finite additive integral in (3.4) is a *bound on the source contribution*; it does not independently bound the signed work $\mathcal{I}_f$ when Y is uncontrolled. Forcing changes the solution and may therefore change g_χ and J_χ. Only the functional form of the stretching estimate is unchanged. That form alone cannot label a history regular or singular; its accumulated value supplies the distinction.

4 The Reported Breakdown and Its Integral Consequence

OpenAI's Theorem 1.1 [3] reports, for every $\nu > 0$, a real-valued smooth force compactly supported in space and time and a solution on $\mathbb{R}^3 \times [0, 1)$ starting from rest, with uniformly bounded kinetic energy and unbounded maximum velocity as $t \uparrow 1$. Corollary 10.6 gives the periodic counterpart on the unit torus, with smooth forcing through time one and no smooth continuation for those same data and force. The periodic version is the one used below, so the domains and positive viscosity match Proposition 3.1.

Corollary 4.1 (Divergence for the reported periodic family)**.** *Accept the periodic breakdown asserted in [3, Corollary 10.6], and denote its velocity by U. For every fixed unit direction χ,*

$$J_\chi(U; 1) = \lim_{t\uparrow 1} \int_0^t \|P_\chi(\nabla \times U(s))\|_{L^2(\Omega_1)}^4 \,\mathrm{d}s = \infty. \tag{4.1}$$

Proof. If the integral were finite, Proposition 3.1 would give a smooth continuation through time one. By uniqueness, that continuation agrees with U on $[0, 1)$; smoothness on the compact slab $\Omega_1 \times [0, 1]$ makes its velocity bounded there, contradicting $\limsup_{t\uparrow 1} \|U(t)\|_{L^\infty(\Omega_1)} = \infty$ in [3, Corollary 10.6]. The argument applies separately to each fixed direction. □

The consequence is stronger than observing that the available energy estimate does not bound J_χ: under the stated external theorem, there is an actual history for which it diverges. The criterion's hypothesis therefore marks a genuine distinction in the smoothly forced solution class, not merely a

quantity that the present estimation method fails to evaluate. This does not show that the estimate is quantitatively sharp, or exclude sharper estimates on a restricted class. It does rule out universal finiteness of J_χ over a class containing the constructed history.

The results are not contradictory. The unforced matrix application and its conditional consequence do not assert smoothness for all prescribed smooth forces. The external construction uses nonzero forcing, and its solution fails the finite-integral hypothesis even though the force's own contribution is bounded on the time slab. Mathematical admissibility within these equations is the relevant statement; no additional claim of laboratory realization or of nonphysicality is needed.

4.1 What the core scaling does, and does not, identify

The leading-core scales reported in [3, Sec. 2.1], with $\tau = 1 - t$ and fixed $0 < h < 1/100$, are radial length $\ell_r \asymp \tau^{1/2}$, axial length $\ell_z \asymp \tau^{1/2-h}$, and characteristic tangential and axial speed $V \asymp \tau^{-1/2-h}$. A useful dimensional comparison, based on radial derivatives at scale V/ℓ_r, is

$$\begin{aligned} \text{core kinetic-energy scale :} \quad & V^2 \ell_r^2 \ell_z \asymp \tau^{1/2-3h}, \\ \text{core enstrophy-norm scale :} \quad & (V/\ell_r)^2 \ell_r^2 \ell_z \asymp \tau^{-1/2-3h}. \end{aligned} \tag{4.2}$$

The first tends to zero while the second increases. The latter exponent approaches the parabolic self-similar enstrophy exponent $-1/2$ as $h \downarrow 0$. This is the sense in which the leading-core diagnostic is near critical. Equation (4.2) is a scale calculation, not a proved two-sided estimate for the complete corrected field.

The construction illustrates the physical distinction in Section 2: its core combines inward spiralling with axial motion, rather than planar transport alone [3, Fig. 1 and Sec. 2.1]. Both ℓ_r and ℓ_z shrink, but $\ell_r/\ell_z \asymp \tau^h \to 0$, so the concentration becomes increasingly slender. This is relative elongation, not growth of its absolute height. The geometry illustrates one three-dimensional concentration mechanism; the two-component criterion does not require every possible breakdown to have that particular spiral geometry.

The projection requires a separate distinction. Wherever $Y(t) > 0$, define

$$q_\chi(t) = \frac{\|P_\chi \omega(t)\|_2^2}{Y(t)} \in [0,1], \qquad \|P_\chi \omega(t)\|_2^4 = q_\chi(t)^2 Y(t)^2. \tag{4.3}$$

If the total norm had the scale in (4.2) and q_χ were bounded below by a positive constant near the horizon, the integral would have the divergent comparison $\int \tau^{-1-6h}\,\mathrm{d}t$. Neither a total-vorticity scale alone nor bounded kinetic energy establishes that projection hypothesis. Corollary 4.1 needs no such lower bound: it establishes divergence by the continuation implication and the external theorem, but does not assign an asymptotic rate to it.

5 Exact Finite Examples and the Remaining Scope

The finite side also has explicit families. For $k = 2\pi m/L$, m a positive integer, and real amplitude A_0, the unforced periodic shear

$$u(x,t) = (A_0 e^{-\nu k^2 t} \sin(k x_2), 0, 0), \qquad p = 0 \tag{5.1}$$

has no advective term. Its vorticity is $(0, 0, -A_0 k e^{-\nu k^2 t} \cos(kx_2))$ and

$$Y(t) = Y_0 e^{-2\nu k^2 t}, \qquad Y_0 = \frac{A_0^2 k^2 L^3}{2}, \qquad J_\chi(u;T) = \frac{(1-\chi_3^2)^2 Y_0^2}{4\nu k^2}\left(1 - e^{-4\nu k^2 T}\right). \tag{5.2}$$

This verifies finiteness for each member even as $T \to \infty$. It is one parameter-dependent certificate for a family, not a separate test of each member. Planar velocities $u = (v_1(x_1, x_2, t), v_2(x_1, x_2, t), 0)$ have $P_{e_3}\omega = 0$, giving a zero-block example. A general three-component velocity independent of x_3 need not have this property: its third velocity component can contribute horizontal vorticity.

Table 5.1: Selected integral outcomes. All entries use positive viscosity and the stated fixed projection.

Case and input class	Horizon and integral	Basis
Unforced shear (5.1), Ω_L, any fixed χ	Finite for every finite T, and for the infinite-horizon total	Exact expression (5.2)
Planar unforced flow, Ω_L, $\chi = e_3$	$J_{e_3} = 0$ on each smooth interval	Identically zero projected vorticity
Reported forced family, Ω_1, any fixed χ	$J_\chi(U;1) = \infty$	Corollary 4.1, conditional on the external breakdown theorem
Arbitrary smooth unforced periodic data	Class-wide finiteness on every finite horizon remains open	Not decided by the particular families above

For numerical cases, a rigorous enclosure of the field can supply an upper certificate. If an exact solution is validated on $[0, T]$ and $\|P_\chi(\omega - \nabla \times v)\|_2 \le \varepsilon_\chi(t)$, then

$$J_\chi(u;T) \le \int_0^T \left(\|P_\chi(\nabla \times v(t))\|_2 + \varepsilon_\chi(t)\right)^4 \, \mathrm{d}t. \tag{5.3}$$

This elementary enclosure is compatible with the a posteriori existence and robustness approach of Chernyshenko, Constantin, Robinson, and Titi [5]; it does not itself validate the approximation. No new numerical certificate is reported here. Failure to obtain an upper certificate is not a proof of divergence.

For the unforced periodic class, finiteness of J_χ for every smooth divergence-free initial field and every finite horizon would, by Proposition 3.1, imply global smoothness. Conversely, a global smooth periodic solution has finite J_χ on every compact time slab. With the same data and periodicity assumptions, this is the scope of alternative (B) in the classical formulation [7]. The forced example does not decide it. The present estimate also requires $\nu > 0$; the coefficient ν^{-3} provides no criterion at $\nu = 0$, so Euler constructions are outside this particular comparison.

6 Conclusion

The comparison has a definite result. The continuation coefficient selected in the matrix-of-error application is the $L_t^4 L_x^2$ two-component criterion of established regularity mathematics. Its fourth power and viscosity dependence follow from estimating stretching against diffusion, while the usual energy bound alone does not supply its cumulative control. This explains the condition analytically, without requiring the physical hypothesis that initially motivated the follow-up.

Adding a smooth force preserves the blockwise stretching estimate and adds an explicitly controlled inhomogeneous term to the envelope. It does not preserve the trajectory or guarantee that the trajectory-dependent integral remains finite. Accepting the reported periodic breakdown, that integral must diverge for every fixed two-component projection at the singular horizon. The constructed family consequently demonstrates failure of the criterion's hypothesis in an admissible forced setting, not failure of the conditional criterion or contradiction of its unforced applications.

The contribution of this note is this explicit connection: a known continuation condition, its matrix-of-error realization, and a reported singular family can be compared using the same accumulated quantity, with the change of input class and the force contribution fully visible. The finite examples and the qualified core calculation clarify the comparison without classifying every history. No conclusion about class-wide unforced smoothness follows from the forced construction, and none is asserted here.

Preparation and Provenance

The author originated the nonlinear-calculus and matrix-of-error proposals and directed the preceding research and this follow-up comparison. The author's broader physical hypothesis motivated the investigation; the present version concentrates on mathematical statements that do not depend on that hypothesis. Fable 5.1 proposed the short-note structure and the use of the existing criterion attribution, scaling explanation, and forced comparison as its organizing subjects. At the author's direction, GPT 6 reworked version 2, checked the relevant primary-source statements, supplied the explicit periodic block estimate, and assembled this expository version. No new experiments or formal proof builds were performed. The external-result consequence is conditional on the reported theorem, not a claim of an independent complete audit of its proof or formalization.

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